

THE GEOLOGY OF ETHIOPIA AND THE HORN

Outline

- ✓ *The Geologic Processes: Endogenic and Exogenic Forces*
- ✓ *The Geological Time Scale and Age Dating Techniques*
- ✓ *Geological Processes & the resulting Landforms of Ethiopia and the Horn*
- ✓ *Rock and Mineral Resources of Ethiopia*

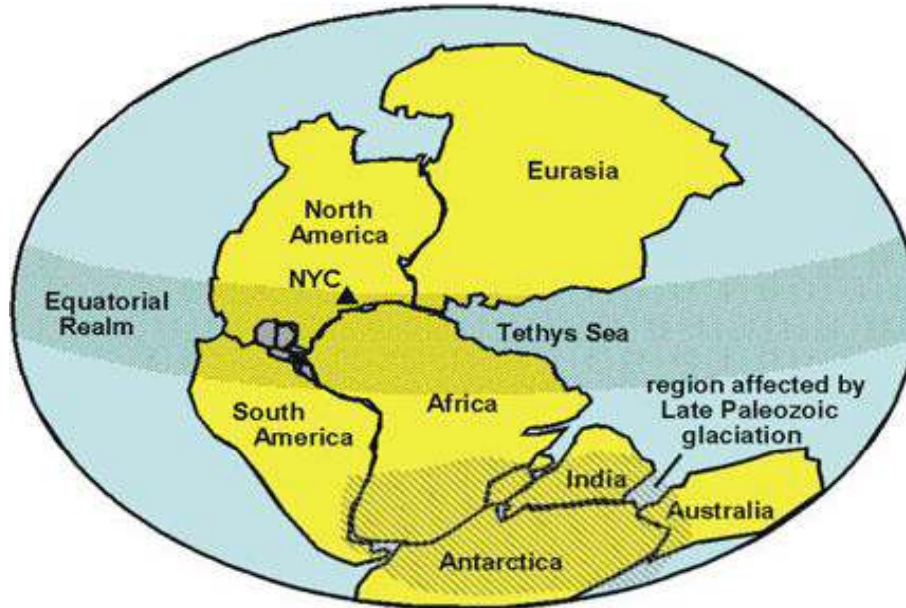
Introduction

- **Geology** is science of earth that studies;
 - ✓ the evolution,
 - ✓ the materials composition and
 - ✓ the processes acting upon Earth
- It studies how earth's materials, structures, processes and organisms have changed over time.

Intro...

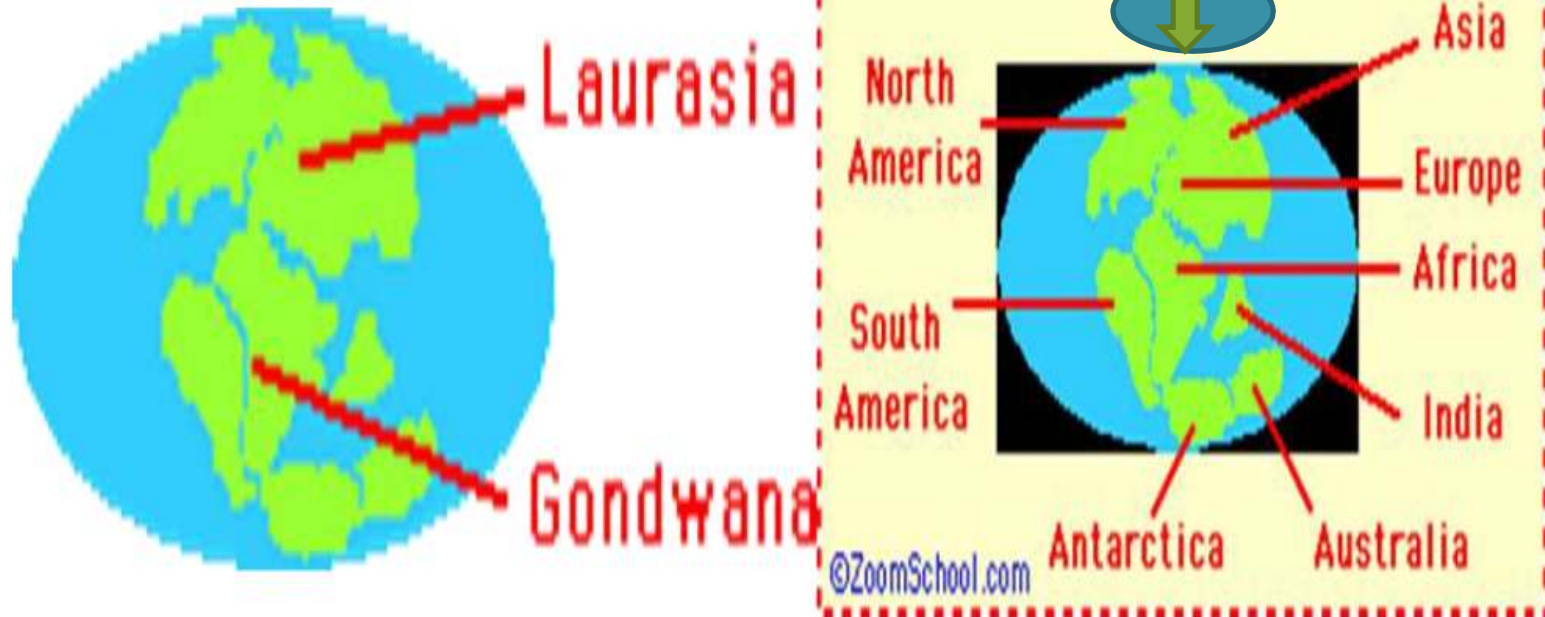
- The earth's continents were once bunched up together into a single huge continent called **Pangaea**.

The Pangaea



- **Pangaea** split into *Gondwanaland* and *Laurasia* and later into smaller fragments over the last million years.
- The two larger landmasses were separated by an elongated sea known as **Tethys sea**

Broken Pangaea



➤ Two theories describe about this issue: ***Continental Drift*** and ***Plate Tectonics theory***.

The Continental drift theory

- ✓ Continental drift is the horizontal motion of the continents.

¶ *The theory of the continental drift was formulated by Alfred Wegner in 1912. Wegner proposed that about 200 million years ago, all the present day continents joined together and formed a mega continent called PANGAEA (meaning all the lands, Greek). Pangaea was surrounded by a mega water body called PANTHALASSA.*

- ✓ Pangaea broke into Laurasia and Gondwanaland forming the northern and southern continents respectively.

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✓ **Laurasia** is further broken down into;

- ✓ *North America,*
- ✓ *Europe,*
- ✓ *Asia*
- ✓ *Green Island*

✓ **Gondwanaland** fragmented into;

- ✓ *Africa,*
- ✓ *South America,*
- ✓ *Australia,*
- ✓ *Antarctica,*
- ✓ *Indian sub-continent,*
- ✓ *Arabian Peninsula*
- ✓ *Madagascar Island*

What were the evidences forward by Alfred Wegner in support of the Continental drift theory?

- ✓ ***The Matching of shorelines of Continents (Jig-Saw-Fit):*** *The shorelines of Africa and South America facing each other have a remarkable and unmistakable match like a jig-saw puzzle.*
- ✓ ***Geologic Similarity:*** *rocks of same age in different areas; example: basement complex rock of Gabon (Africa) and Brazil.*
- ✓ ***Paleoclimatic (ancient climate evidences):*** *similar climate*
- ✓ ***Similar orogenic Belts:*** *the same age mountain*
- ✓ ***Distribution of Fossils:*** *availability of fossil of plants and animals with the same age*
- ✓ ***Paleomagnetism:*** *the same time rock formation from magnatic minerals*

The Geologic Processes: Endogenic and Exogenic Forces

✓ **Geomorphic processes** are forces that bringing changes in the configuration of the surface of the earth.

1) **Endogenic (internal) processes:** are forces that come from the inside part of the earth/operating beneath the earth surface.

Example: Volcanic activity, earth quake, tectonic processes (folding, faulting, orogenesis (mountain building), and epeirogenesis (slow rising and sinking of the landmass)).

2) **Exogenic (external) processes:** are forces that act on the surface of the earth.

Example: weathering, mass wasting, erosion and deposition.

➤ In our surrounding various land features are formed b/c of the movement of the earth. There are two types of earth movements.

The Geological Time Scale and Age Dating Techniques

- Using d/t techniques scientists have been able to decide the age of the earth.
- The general history of the earth is described by **what is known as the geological time scale**.
- It shows the age of the earth and accordingly, the age of the earth is known to be **4.6 billion** years old.
- The scale is divided into: Era, Period and Epoch.
- The longest units are called **eons**, and progressively shorter subdivisions are *eras, periods, and epochs*.

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- ✓ **Eon** is one of the principal divisions in the geological time scale
- ✓ An **epoch** is the smallest unit of time on the scale and encompasses a period of millions of years.
- ✓ There are four major era

-Precambrian – the oldest era

-Palaeozoic,

- Mesozoic

- Cenozoic- the recent era

The Geological time scale

Era	Period	Began (million years)	End	Major events(Million years ago)
4. Cenozoic	Quaternary	1.6	present	Major glaciers in North America and Europe (1.5)
	Tertiary	70	1.6	Rocky Mountains (65), individual continents take shape.
3. Mesozoic	Cretaceous	146	70	Dinosaurs extinct (65), western interior seaway and marine reptiles (144 – 65)
	Jurassic	208	146	Pangaea (one land mass) begins to break up (200)
	Triassic	225	208	First mammals and dinosaurs

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2. Paleozoic	permian	290	225	Greatest extinction on Earth (245)
	Pennsylvanian	322	290	Greatest extinction on Earth (245)
	Mississippian	362	322	Coal-forming forests
	Devonian	408	362	First land animals and first forests (408)
	Silurian	439	408	Life invades land
	Ordovician	510	439	First fish appeared
	Cambria	600	510	Great diversity of marine invertebrates
1. Precambrian	Proterozoic	2,500	600	Marine fossil invertebrates (600)
	Archean	4,500	2,500	Earliest fossils recorded (3,500), earliest rock formation (4,000)

Age Dating Techniques

- There are two techniques of knowing the age of rocks:

A. Relative Dating

- ✓ There are two ways to know the relative age of a rock:
 - i) By studying *fossils found in the rock*: it is possible to determine during which period the rock was formed.
 - ii) By studying *layer of the rock* which shows younger rocks are on top of older rocks.

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B. Absolute Dating: known as Radiometric techniques because it uses radioactive element like **uranium** (U) & **thorium** (Th) decay to form different elements/isotopes of the same element.

- ✓ i. **Carbon-14 Technique:** Upon the organism's death, carbon-14 begins to disintegrate at a known rate, and no further replacement of carbon from atmospheric carbon dioxide can take place.
- ✓ Carbon-14 has half-life of 5730 years.
- ✓ ii. **Potassium-Argon Technique:** is widely used for dating rocks.

Geological Processes and the resulting Landforms of Ethiopia and the Horn

The Precambrian Era Geologic Processes (4.5 billion - 600 million yrs).

- It is the **oldest** and **longest** geological era covering 5/6th of the earth's history.
- The major geologic event of the Precambrian era was **Orogenesis**.
- Formation of folded mountain ranges in the Horn of Africa.
- In most parts of Ethiopia rocks belong to this era were found **beneath all others**, forming **basement complex** or **crystalline complex rocks**.

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- ✓ The Precambrian rocks are overlaid by recent rock formations.
- ✓ However, due to continuous **denudation** (wearing away) of the covering rocks, these rocks are found currently **outcropping to the surface** in *Western lowlands, parts of northern and central Tigray*.
 - a) In the western Part: Gambella, Benishangul-Gumuz (Metekel and Asossa), western Gojjam, western Wellega, Illuababora, and Abay gorge.
 - b) In the southern Part: Guji, southern Omo, and parts of southern Bale and Borena.
 - c) In the eastern part: Eastern Hararghe.

The Paleozoic Era Geologic Processes (600 million - 225 million years ago)

- It was often known as an era of **Invertebrates** .
- The major geological process was **denudation** and **peneplanation**.
- Horn of Africa was subjected to complete denudation and peniplanation.
- Gondwanaland was highly eroded.

The Mesozoic Era Geologic Processes (225-70 million years ago)

- It is the era of Reptiles.
- It was an Era of alternate slows sinking and rising (epeirogenesis) of the landmass.
- It was an era when the **Dinosaurs disappeared**.
- Three major sedimentary formations were observed during this era.
 - ✓ *Triassic period- Adigrat sandstone was formed*
 - ✓ *Jurassic period- Hintalo limestone*
 - ✓ *Cretaceous period- Upper sandstone*

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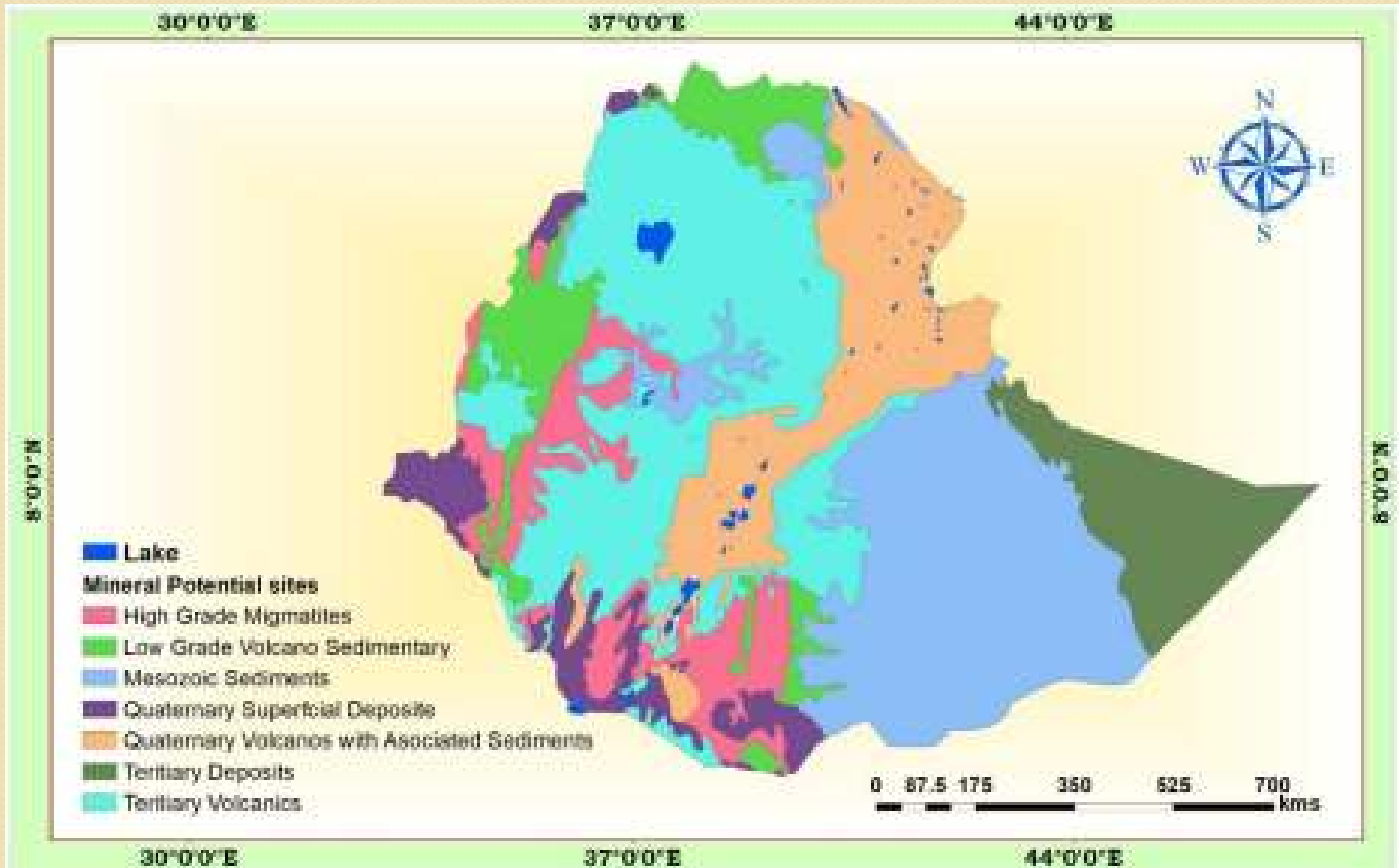


Figure 2. 1. Geological map of Ethiopia Source: CSA Shapefile, 2012

The Cenozoic Era Geologic Processes (70million years ago - Present)

- It is the **most recent** of the geologic eras and it is the era of modern man.
- The present day landmass of Ethiopia and the Horn of Africa are the results of Endogenic and Exogenic forces of this geological era.
- ✓ The major geologic activities are:
 - a) *Uplifting of the Arabo-Ethiopian landmass and outpouring of huge quantity of lava.*
 - b) *Formation of the Rift Valley.*
 - c) *Quaternary volcanism and deposition*

The Formation of the Rift Valley

- Rift valley is an elongated trough/depression bounded by faults.
- Its formation is related with the theory of plate tectonics.
- The lateral movement of the crust in opposite directions producing **tensional forces** caused parallel fractures or faults on the sides of the up-arched swell.

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- ✓ As the tension widened the fractures, the central part of the landmass collapsed to form an extensive structural depression known as the

Rift Valley.

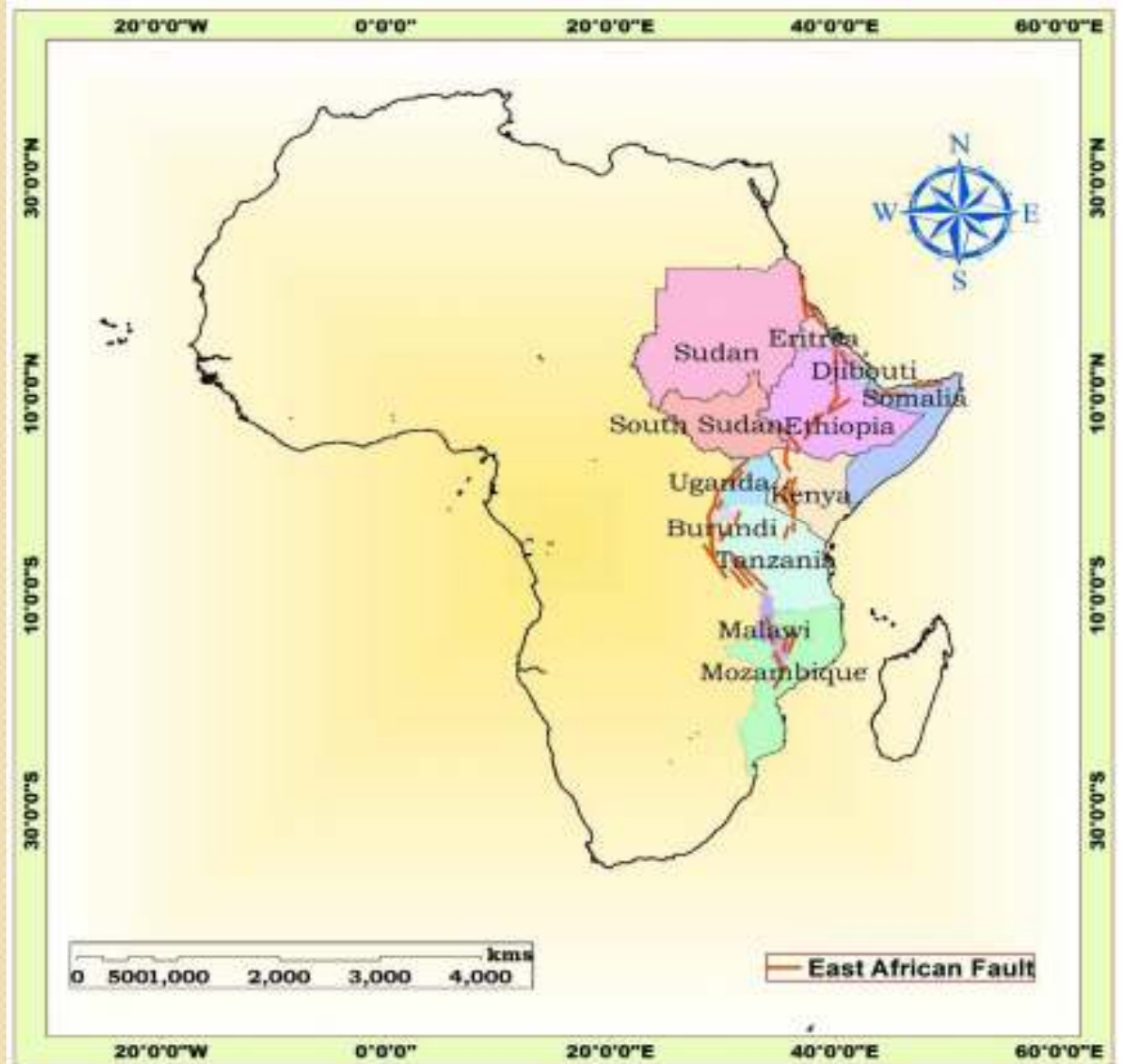


Fig. The Ethiopian and East African Rift Valley systems.

Source: Africa Map Library

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The Spatial Extent of the Rift Valley

- It is part of the Great East African Rift system that extends from **Palestine-Jordan** in the north to **Malawi-Mozambique** in the south,
- ✓ Cover distance of about 7,200 kilometers.
- ✓ 5,600 kilometers is in Africa, and
- ✓ 1,700 kilometers in Eritrea and Ethiopia.
- ✓ the widest part of the Rift Valley is the Afar Triangle (200-300 km).

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- The Rift Valley region of Ethiopian is **the most unstable part of the country**. There are numerous hot springs, active volcanoes, geysers, and frequent earthquakes.
- The formation of the Rift Valley has the following structural (physiographic) effects:
 - It divides the Ethiopian Plateau into north western and southwestern highlands.
 - It separates the Arabian landmass from African landmass.

Quaternary Deposition

- During the Quaternary period, the *earth experienced a marked climatic change.*
- In the middle and high latitude areas and the time of the ‘’**Pluvial Rains**’’ in Africa.
- The heavy Pluvial Rains eroded the Ethiopian plateau and the eroded materials were deposited in the Rift Valley lakes.

Rock and Mineral Resources of Ethiopia

- They are naturally occurring and inorganic substance that are useful for human beings in many ways:
 - ✓ serve as raw materials for manufacturing,
 - ✓ source of energy (coal, petroleum, and natural gas),
 - ✓ for ornamental purpose (gold, diamond,),
- Ethiopia is endowed with a wide variety of minerals and rocks, some of which are available in large quantities and are of excellent quality.
- Example :Gold, Tantalum, Potash, Platinum etc.
- However, presently mineral production from Ethiopia has been negligible by World standards.

Mineral Potential Sites of Ethiopia

- ✓ According to the Ethiopian geological survey, the formations that host most mineral potentials of Ethiopia includes 3 major greenstone belts and other formations. These are:
 1. **The Western and South-western-greenstone belt:**
 - ✓ They contain d/t minerals like gold occurrences (Dul, Tulu-Kape, Oda-Godere Akobo, Baruda, Bekuji-Motish and Kalaj);
 - ✓ Yubdo Platinum,
 - ✓ Base metals of Azali Akendeyu,
 - ✓ Abetselo and Kata; Fakushu Molybdenite and
 - ✓ the iron deposits of Bikilal, Chago, Gordana and Korre, Benshagul-Gumuz-Marble, Akobo and Asosa placer gold deposits and etc.

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2. The Southern greenstone belt:

- ✓ It is known as the Adola belt,
- ✓ which comprises the primary gold deposits and occurrences of Lega-dembi, Sakaro, Wellena, Kumudu, Megado-Serdo, Dawa Digati, Moyale and Ababa River;

3. The Northern greenstone belt(Tigray):

- ✓ This belt comprises of the primary gold occurrences of Terakemti, Adi-Zeresenay, and Nirague.